2.2.3] Logistic Regression

Definition

Logistic Regression is a supervised learning algorithm used for **classification problems**.

It predicts the **probability** of a categorical outcome (usually 0 or 1).

Example: Pass/Fail, Yes/No, Spam/Not Spam, Disease/No Disease.

Need of Logistic Regression

Linear Regression gives output values from −∞ to +∞

But in classification:

- Output should be between **0 and 1**
- Representing probability

Hence Logistic Regression uses **Sigmoid Function**.

Sigmoid Function

Definition -

The **Sigmoid Function** is a mathematical function used in **Logistic Regression** to convert any real value into a probability value between **0 and 1**.

It is also called the **Logistic Function**.

Logistic Regression Formula

First, we calculate:

Z=b0 + b1 X1 + b2 X2 + ...+ bn Xn

Then apply Sigmoid function:

$$P(Y = 1) = \frac{1}{1 + e^{-Z}}$$

Where:

- P(Y=1) = Probability of class 1
- b0 = Intercept
- b1,b2..bn = Coefficients

- X1,X2,…. Xn = Independent variables

Example

Problem Statement

Predict whether a student will **Pass (1)** or **Fail (0)** based on:

- Study Hours
- Attendance (%)

This is a **Classification Problem** because:

- Output is categorical (Pass/Fail)

Sample Dataset

Study Hours	Attendance	Result
1	60	0
2	65	0
3	70	0
4	75	1
5	80	1
6	85	1
7	90	1

Where:

- 0 = Fail
- 1 = Pass

. Python Code

```
# Import libraries
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Create dataset
data = {
    'StudyHours': [1,2,3,4,5,6,7],
    'Attendance': [60,65,70,75,80,85,90],
    'Result': [0,0,0,1,1,1,1]
}

df = pd.DataFrame(data)

# Independent variables
X = df[['StudyHours', 'Attendance']]

# Dependent variable
y = df['Result']

# Create model
model = LogisticRegression()

# Train model
model.fit(X, y)
```

```
# Predictions
predictions = model.predict(X)

# Probability predictions
probabilities = model.predict_proba(X)

# Print results
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
print("Predictions:", predictions)
print("Accuracy:", accuracy_score(y, predictions))
print("Confusion Matrix:\n", confusion_matrix(y, predictions))
```

✓ Step 3: Sample Output (Approximate)

```
Intercept: [-10.50]
Coefficients: [[1.20  0.15]]

Predictions: [0 0 0 1 1 1 1]

Accuracy: 1.0

Confusion Matrix:
[[3 0]
 [0 4]]
```

Output Explanation

Logistic Regression Equation

Unlike Linear Regression, Logistic Regression uses **Sigmoid Function:**

$$P(Y=1) = \frac{1}{1+e^{-(b0+b1X1+b2X2)}}$$

From output:

$$Z = -10.50 + 1.20(StudyHours) + 0.15(Attendance)$$

Then probability:

$$P = \frac{1}{1+e^{-Z}}$$

Meaning of Coefficients

◆ Study Hours (1.20)

If study hours increase by 1 unit, probability of passing increases.

◆ Attendance (0.15)

If attendance increases by 1%, probability of passing increases.

Positive coefficient → Increases chance of Pass
Negative coefficient → Decreases chance

Predictions

```
[0 0 0 1 1 1 1]
```

Model correctly classified:

- First 3 students as Fail
- Next 4 students as Pass

Accuracy

```
Accuracy: 1.0
```

Accuracy = 100%

Means:
Model correctly predicted all students.

Confusion Matrix

```
[[3 0]
 [0 4]]
```

	Predicted Fail	Predicted Pass
Actual Fail	3 (Correct)	0
Actual Pass	0	4 (Correct)

No misclassification.

Why This Is Logistic Regression?

Because:

- Output variable is categorical (0 or 1)
- Uses sigmoid function
- Predicts probability

Important Viva Questions

1. Why don't we use Linear Regression for classification?
2. What is Sigmoid Function?
3. What is decision boundary?
4. Why output is between 0 and 1?
5. What is confusion matrix?

✓ Step 7: Difference Between Linear & Logistic Regression

Linear Regression	Logistic Regression
Predicts continuous value	Predicts category
Uses straight line	Uses sigmoid curve
Output any value	Output between 0 and 1
Used for regression	Used for classification

★ 10-Mark Exam Writing Format

1. Define Logistic Regression
2. Write sigmoid formula
3. Give example
4. Explain coefficients
5. Explain confusion matrix
6. Write conclusion

Conclusion

Logistic Regression is used for binary classification problems.
It predicts probability using sigmoid function and classifies data based on threshold (usually 0.5).